MEDICO ENGINEER

MEDICO ENGINEER

Your Free MDCAT Companion

Serving MDCAT students for 4 years — completely free of cost

Prepare smarter with trusted, high-quality MDCAT resources designed to help students practice, revise, and improve confidently.

We Provide

Free High-Quality Practice Tests

Chapter-wise and topic-wise MDCAT practice tests for strong exam preparation.

MDCAT Books

Useful books and study resources for Biology, Chemistry, Physics, English, and Logical Reasoning.

MDCAT Material

Organized MDCAT preparation material for students.

Notes & Revision Guides

Easy notes for quick revision and concept clarity.

Top Academies Material

Selected material from well-known academies to support your MDCAT preparation.

Access Free Testing Service

Join Our WhatsApp Channel

Scan the QR Code to Join

Connect With Us

Facebook Page: Medico Engineer

WhatsApp: 03700113837

Website: Medicoengineer.com

Medico Engineer

Free MDCAT Preparation Platform for Every Student

Prepare • Practice • Improve • Succeed

SINDH

MDCAT PAST PAPER

2013

FOR MORE INFORMATION VISIT OUR FACEBOOK PAGE
PRESTIGIOUS TESTING SERVICE-PTS

COMPILED BY: RAM LAL VALASAI

For Study Material Visit: Medicoengineer.com

NTS past paper 2013
ENGLISH

Complete the sentences by choosing the most appropriate notion, from the given lettered choices (A to E) below each.

1. I_____ have to get up early tomorrow morning.
A. shall
B. would
C. had
D. could
E. am

2. He says he is a_____ but he can't play the piano or any other instrument and he can't sing.
A musician
B magician
C. physician
D. dietician
E technician

Identify the word or phrase that needs to be changed for the sentence to be correct:

3. **The** secretary **must** type **these letters**
A B C
before **lunch**. **No error**
D D

4. **I** go by the **post office** every morning
on
A B
C my way **too** work. **No error**
D E

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters.

5. GRIEVE:
A. disturb
B. hurt
C. dirge
D. feel sad
E. rejoice

6. BLAZE:
A. quench
B. burn
C. rage
D. shine
E. flame

Choose the word most similar in meaning to the capitalized ones.

7. GENTLE:
A. rough
B. expert
C. heartless
D. calm
E wicked

8. SQUASH:
A. squeeze
B. beat
C. evolution
D. pace
E. rapidity

Questions 9-10 are based on the following passage.

However, it must be recognized that science has its limitations. Its methods apply only to those things which can be observed, measured, and treated mathematically. It has nothing to do with values - save those of truth and accuracy. It has nothing to do with happiness, goodness, beauty, courage. adventure, justice, altruism, friendship, love of family, love of country. Yet all these values enter into a man's conception of what is the good personal life within a good society. It is possible for honest and intelligent men to differ profoundly on the nature of these values and their respective degrees of

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000–2023
For Study Material Visit: Medicoengineer.com

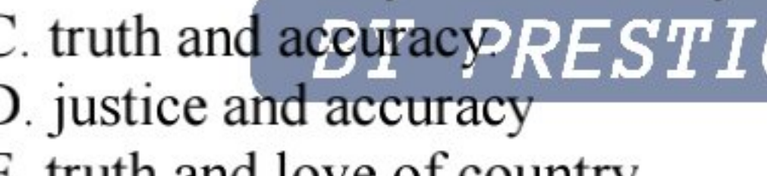
importance. Hence the contrast between the modern worla's command of material things and its tragic failure to organize a harmonious world society.

9. It can be inferred from the passage that the author thinks:
A. science to be the necessary element on life
B. science brings happiness, goodness, beauty, adventure, justice. altruism, friendship, love and love of country to a man's life
C. science has successfully brought a balance between world's command of material things and a harmonious world society
D. a good personal life can be achieved by recog nature of values and their degree of importance
E. the limitations of science are negligible.

10. According to the paragraph science applies certain values, which of the following describes these values?
A. truth and justice
B. love of country and accuracy
C. truth and accuracy
D. justice and accuracy
E. truth and love of country

PHYSICS

11. Two blocks, X and Y, of masses mand 2 m respectively, are accelerated along a smooth horizontal surface by a force F applied to block X, as shown in the diagram.



What is the magnitude of the force exerted by block Yon block X during this acceleration?
A. O
B.F/3
C.F/2
D. 2F/3
E.F

12. A box of mass m = 6 kg slides with speed v = 4 m/s across a frictionless floor. It suddenly explodes into two pieces. One piece, with mass m_1 = 2 kg moves in the same direction with speed V_1= 8 m/s. Find the velocity of the second piece.
A. 2 m/s
B. 4 m//s
C. 8 m/s
D. 9 m/s
E. 11 m/s

13. A generator of e.m.f. 80 V has an internal resistance of 0.04Ω. If its terminal voltage is 75 V, determine the current.
A. 125 A
B.135 A
C. 145 A
D. 155 A
E. 165 A

14. A 4 cm high object is located 10 cm from the converging lens, whose focal length is 20 cm. The image so formed will be:
A. Virtual
B. Erect
C. Real
D. Inverted
E. Both A and B

15. A rotating wheel of radius 0.5 m has an angular velocity of 5 rad/s at some instant and 10 rad/s after 5s find the angular acceleration of a point on its rim.
A. 1 rad/S^2

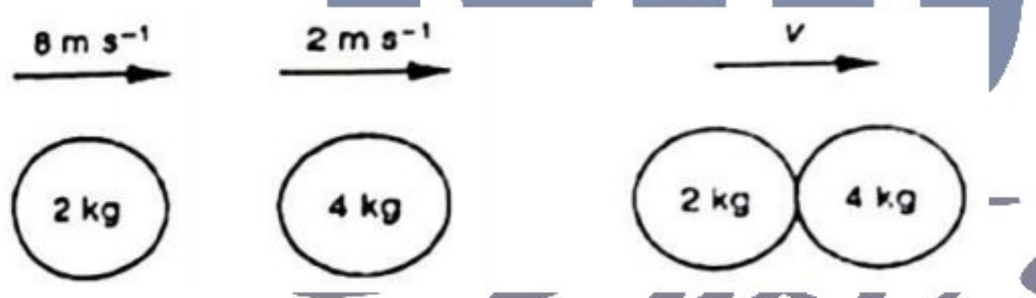
MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

For Study Material Visit: Medicoengineer.com

B. 3 rad S^2
C. 5 rad/S^2
D. 7 rad S^2
E. 9 rad S^2

16. A block of mass 50 kg is pulled on a frictionless floor by a force of 210 N directed at 30° to the horizontal. If the block moves 3.0 m, what is the work done on It by the applied force.
A. $115\sqrt{2}$J
B. $215\sqrt{2}$ J
C. $315\sqrt{2}$ J
D. $415\sqrt{2}$ J
E. $515\sqrt{2}$ J

17. A ball of mass 2 kg traveling at 8 ms^{-1} strikes a ball of mass 4 kg traveling at 2 ms^{-1}. Both balls are moving along the same straight line as shown.



After collision, both balls move at the same velocity v. What is the magnitude of the velocity v?
A.4ms^{-1}
B. 5ms^{-1}
C. 6ms^{-1}
D. 8ms^{-1}
E. 10ms^{-1}

18. A shot leaves a gun at the rate of 160 m/s. Calculate the greatest distance to which it could be projected. (Take g=10m/s^2) .
A. 2460 m
B. 2560 m
C. 2680 m
D. 2760 m
E. 2860 m

19. On the ground, the gravitational force on a satellite Is W. What is the gravitational force on the satellite when at a height R/50, where R is the radius of the Earth?
A 1.04 W
B. 1.02 W
C. 0.98 N
D. 0.96 W
E. 2.13 W

20. When the aircraft Concorde is moving in a horizontal plane at a constant speed of 650ms^-1. its turning circle has a radius of 80 km. What is the ratio of the centripetal force to the weight of the aircraft? (g=9.8 m/s^2)
A. $8.3x10^4$
B. 0.54
C. 1.9
D. 52
E. 540

21. The amount of heat at constant volume is called as:
A. Internal energy
B. Enthalpy
C. Entropy
D. Temperature
E. Pressure

22. A parallel beam of white light is incident normally on a diffraction grating. It is noted that the second-order and third-order spectra partially overlap. Which wavelength in the third-order spectrum appears at the same angle as the wavelength of 600 nm in the second-order spectrum?
A. 300 nm
B. 400 nm
C. 600 nm
D. 900 nm.
E. 950 nm

MDCAT Papers
BY PRESTIGIOUS TESTING SERVICE
2000–2023

For Study Material Visit: Medicoengineer.com

23. If the frequency of a pendulum is four times greater on an unknown planet than it is on earth then the gravitational cons on that planet is;
A. 16 times greater
B. 4 times greater
C. 4 times lower
D. 16 times lower
E 24 times lower

24. A submarine sends out a sonar signal (sound wave)" in a direction directly downward. It takes 2.3 s for the sound wave to from the submarine to the ocean bottom and back to the submarine. How high (approx) up from the ocean to submarine? (The speed of sound in water is 1,490 m/s.)
A. 1,700 m
B. 3,000 m
C. 5,000 m
D. 9,000 m
E. It cannot be determined from the information given

25. A 40 kg block is resting at a height of 5 m off the ground. If the block is released and falls to the ground, what is its total energy at a height of 2 m? (g = 10 m/s²)
A. 0J
B. 400J
C. 2 kJ
D. 6 kJ
E. It cannot be determined from the information given.

26. Gamma (γ) ray can produce ionization in which of the following way/s?
I. It may lose all its energy in a single encounter with electron of an atom (Photoelectric effect).
II. It may lose only a part of its energy in an encounter (Compton effect).
III. Very few of very high energy V ray photons may impinge directly on heavy nuclei, be stopped and annihilated giving rise to electron- positron pairs (The materialization of energy).
A. I only
B. II only
C. III only
D. I and III only
E. I, II and III

27. The Internal energy of an object increase in an adiabatic process Which of the following must be the true regarding this process.
A. The kinetic energy of the system is changing
B. The potential energy of the system is changing
C. Work Is done on the system
D. Heat flows into the system
E. No work is done on the system

28. An electric rod of 2000 watts rating bolls a certain quantity of water in 10 minutes, the heat which is generated for boiling this water is:
A. 8×10^4 Joules
B. 12×10^5 Joules
C. 19×10^5 joules
D. 23×10^5 Joules
E. 37×10^5 Joules

29. A nucleus consists of 19 protons and 20 neutrons. The conventional symbol of this nucleus Is:
A. ${}_{11}Na^{12}$
B. ${}_{19}K^{19}$
C. ${}_{19}K^{39}$
D. ${}_{10}K^{20}$
E. ${}_{11}Na^{12}$

30. The linear magnification produced by a lens is defined as the ratio of the:
I. Size of the image to the size of object

MDCAT Past Papers BY PRESTIGIOUS TESTING SERVICE 2000-2023

For Study Material Visit: Medicoengineer.com

II. Size of the lens to the size of object
III Size of the lens to the size of the Image
A. I only
B. II only
C. III only
D. II and III only
E. I. II and III

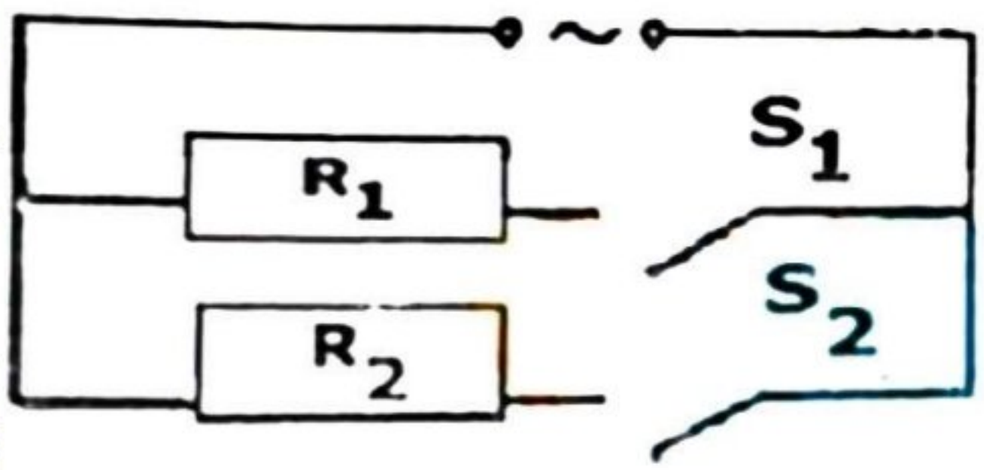
31. The half life of C^{14} is approximately 5,730 years, while the half life of C^{12} is essentially infinite. If the ratio of C^{14} to C^{12} in a certain sample is 25% less than the normal ratio in nature, how old is the sample?
A. Less than 5,730 years
B. Approximately 5,730 years
C. Significantly greater than 5, 730 years, but less than 11,460 years
D. Approximately 11,460 years
E Approximately 15,730 years

32. Which of the following statements is not consistent with Bohr's set of postulates regarding the hydrogen atom model with regard to the emission and absorption of light?
A. Energy levels of the electrons are stable and discrete.
B. An electron emits or absorbs radiation only when making a transition from one energy level to another.
C. To jump from a lower energy to a higher energy, an electron must absorb a photon of precisely the right frequency such that the photon's energy equals the energy difference between the two orbits.
D. To jump from a higher energy to a lower energy, an electron' absorbs a photon of a frequency such that photon's energy is exactly the energy difference between the two orbits.
E. None of the above

33. The temperature of a body at 100°C is increased by A0 as measured on the Celsius scale. How is this temperature change expressed on the Kelvin scale?
A $\Delta\theta$+373
B. $\Delta\theta$ +273
C. $\Delta\theta$ + 100
D. $\Delta\theta$
E. $\Delta\theta$ +212

34.In an astronomical telescope, the distance between objective and eye piece is called:
A. Magnifying Power of the telescope
B. Width of the telescope
C. Length of the telescope
D. Height of the telescope
E. Diameter of the lens of the telescope

35. An electric heater can be represented as two resistors of resistances R_1 and R_2 and two switches S_1 and S_2. The resistance R_2 is greater than that of R_1



Which switches must be closed so that the heater produces the maximum possible power and minimum non-zero power?

	Maximum possible power	Minimum non-zero power
A	S_1 and S_2	S_2
B	S_1 and S_2	S_1
C	S_1	S_2
D	S_2	S_1
E	S_1	S_1

36. Candela is the luminous intensity, in the perpendicular direction of a surface _____square meter of a black body at

MDCAT Past Papers BY PRESTIGIOUS TEAM 2000-2023

For Study Material Visit: Medicoengineer.com

the temperature of freezing platinum under a pressure of 101325 newton per square meter.
A. 1/300000
B. 1/600000
C. 1/900000
D. 1/1200000
E. 1/1500000

37. The work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field which is given by F=4i-3j+2k. is:
A. 45
B. 35
C. 25
D. 15
E. 5

38. A constant force acting on a body of mass 5 kg changes its speed from 2 m/s to 7 m/s in 10 s, the direction of motion of the body remains unchanged. Find the magnitude of the force, (Take g = 10 m/s²)
A. 0.5 N
B. 15N
C. 25N
D. 3.5 N
E 45 N

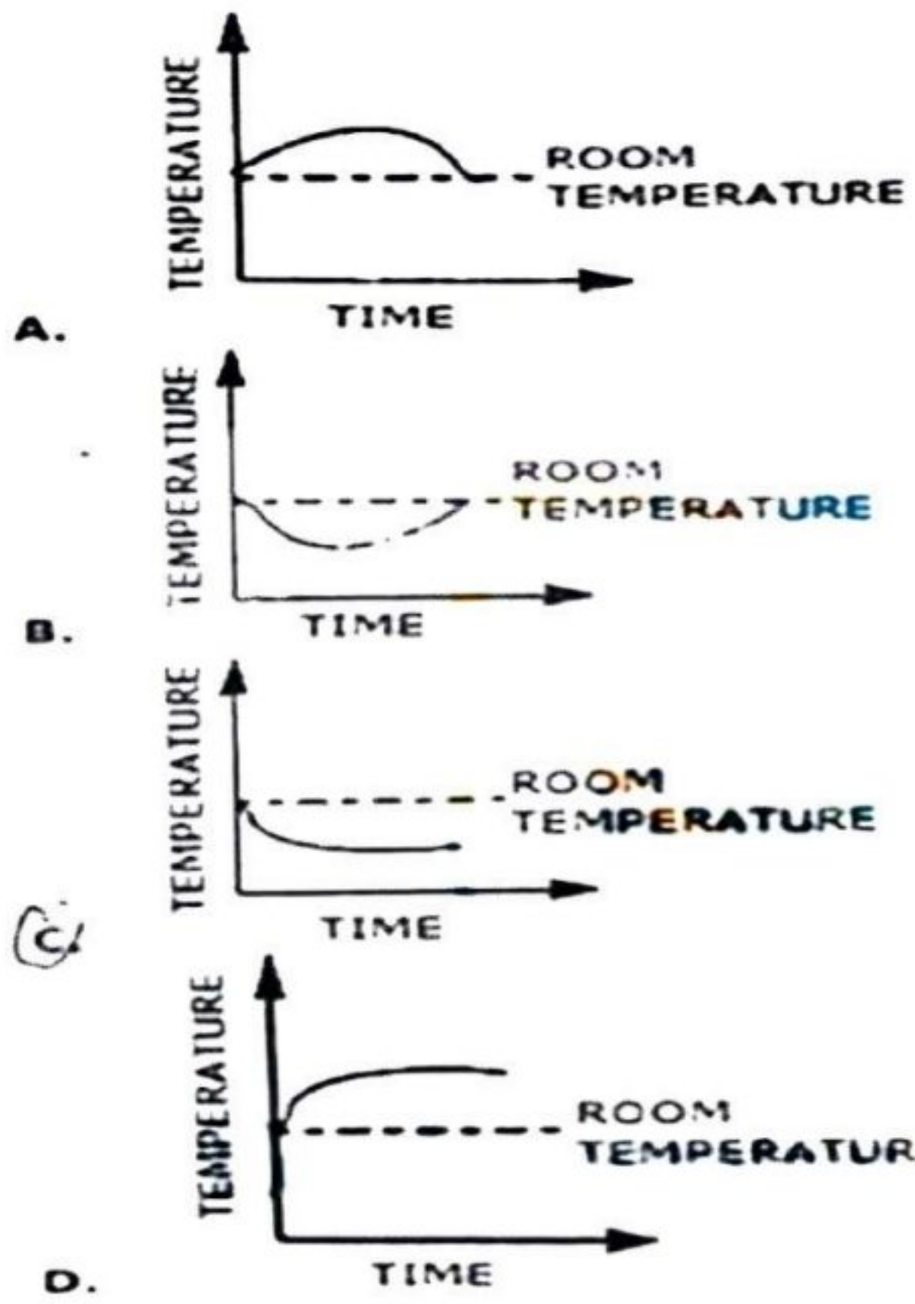
Past
BY PRESTIGIOUS TE
2000-2

39. What force should be applied on a 10 kg body so that it moves down in vacuum with an acceleration of 3 m/s²? (Take g=9.8 m/s²)
A. 42 N
B. 46 N
C 48 N
D. 53 N
E. 58 N

40. A special class of waves which do not need a material medium for their propagation are called:
A. Electric waves
B. Magnetic waves
C.Electromagnetic waves
D.Sound waves
E. Earthquake's shock waves

CHEMISTRY

41. Dissolution of ammonium nitrate in water is an endothermic process. Which of the following graph shows how the temperature alters as the ammonium nitrate is added to water and then the solution is left at room temperature?



42.From a mixture of CO_2, and H_2 gases, CO_2, can be separated by passing the mixture through:
A. water at high temperature
B. water under high pressure
C. cold water
D. acidified water

43. Alkanes having five to seventeen carbon atoms per molecule are:

For Study Material Visit: Medicoengineer.com

A. liquids
B. solids
C. gases
D. semi solid wax

44. Which type of Isomerism depends on distribution of carbon atoms on each side of functional group?
A. Structural Isomerism
B. Functional Isomerism
C. Chain Isomerism
D. Metamerism

45.Non-stoichiometric compounds are formed by:
A. only alkall metals
B. only transition elements
C. only noble gases
D. none of the above

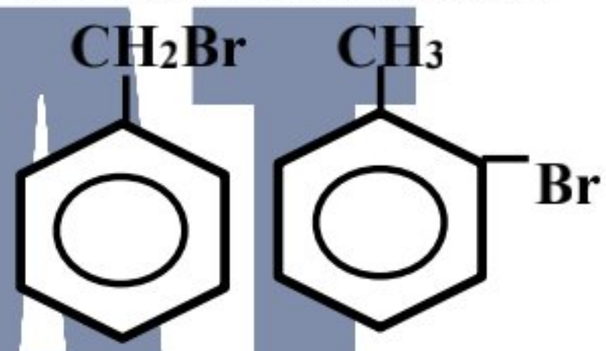
46. Which statement is correct about Nobel gases?
A Their oxidation state is zero
B. They react easily with alkall metals
C. They exist in form of molecules.
D. They are also known as halogens
E. None of the above

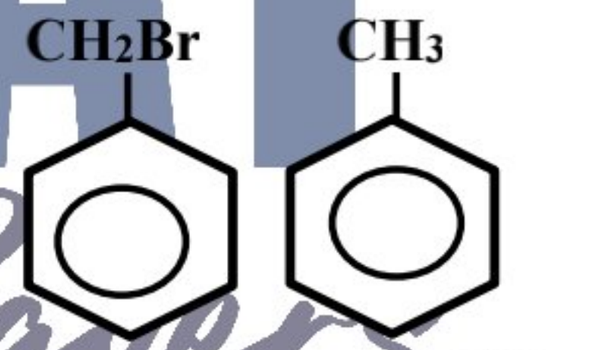
47. Magnesium oxide is used in the making of the lining of blast furnaces. It is extracted from sea water as follows.
Aqueous calcium hydroxide is added to seawater.

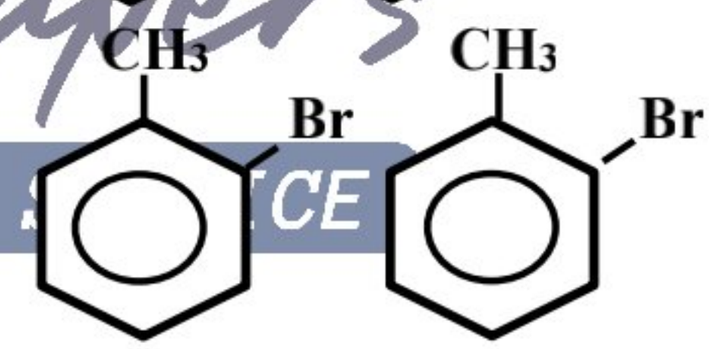
$Ca(OH)2\ (aq) + MgCL_2.\ (aq) \rightarrow Mg(OH)_2\ (s) + CaCl_2.\ (aq)$

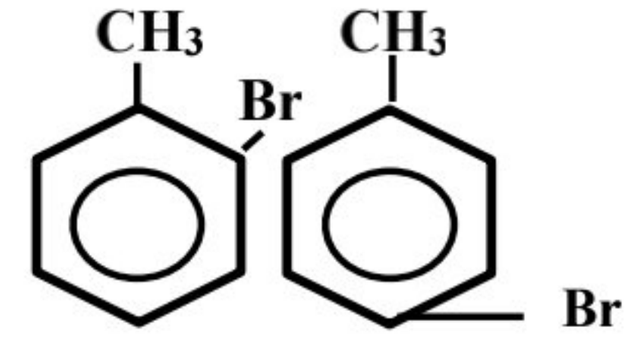
The magnesium hydroxide is then filtered off and roasted.
Which of the following comparisons between calcium and magnesium explains why magnesium hydroxide forms?
A. Magnesium is less electropositive than calcium.
B. Magnesium is lower than calcium in the reactivity series.
C. The enthalpy change of hydration for Mg is less exothermic than for Ca^{+2}
D. The solubility product for $Mg(OH)_2$, is lower than that for $Ca(OH)_2$
E. The magnitude of the lattice energy of $Mg(OH)_2$. is less than that of $Ca(OH)_2$

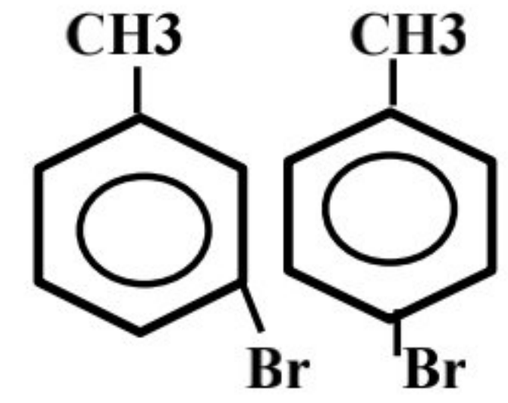
48. When methylbenzene is treated with bromine in the presence of a catalyst, a mixture of two monobromo isomers is formed. What are the structures of these two isomers?

a)


b)


c)


d)


e)


49. The series limit for the Balmer series of hydrogen spectrum occurs

MDCAT Past Papers BY PRESTIGIOUS TESTING SERVICE 2000–2023

For Study Material Visit: Medicoengineer.com

at 3664A°. Calculate Ionization energy of hydrogen atom.

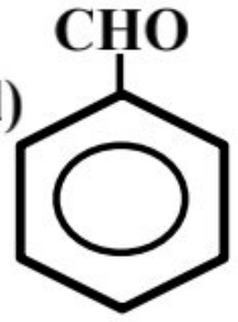
A. 21.7×10^{-19} J
B. 6.626×10^{-34} J
C. 5.425×10^{-19} J
D. 3664×10^{-10} J
E. 3×10^{8} J

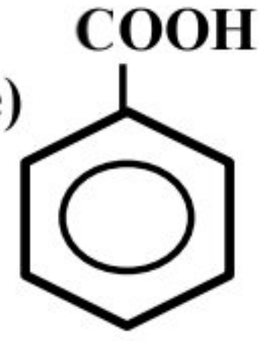
50. Which one of the following formulae represents the organic compound formed when methylbenzene is heated under reflux with alkaline potassium manganate (VII) solution and the mixture then acidified?

a) 

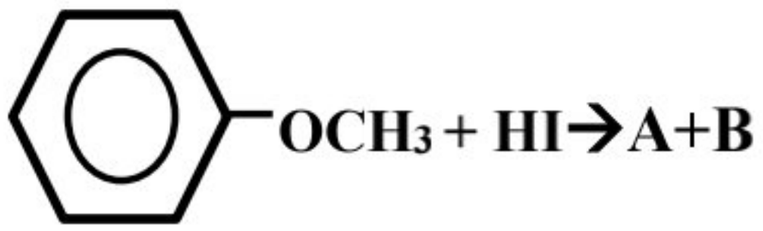
b) 

c) 

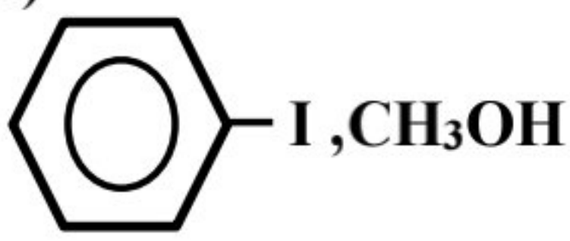
d) 

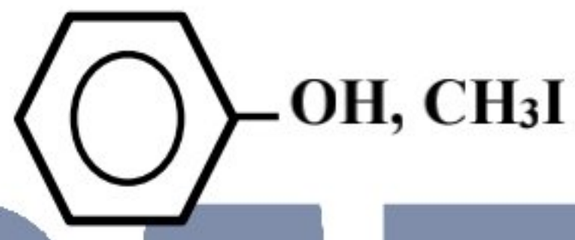
e) 

a)

51. 

A & B are?

a) 

b) 

c) 

d) 

52. The false statement about lithium is:

A. It is softer than other alkali metals
B. It is least reactive
C. It possesses higher melting and boiling points
D. It forms chloride which is soluble in alcohol

53. Diamond and Graphite:

A. are isotopes
B. are Isomers
C. are allotropes
D. have the same structure
E. are equally hard

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

For Study Material Visit: Medicoengineer.com

54. Borax exists In nature as:
A. $Na_2B_40_7 .10H_20$
B. Na $B_40_7.7H_2O$
C. $Na_2B_40_7.5H_20$
D.$Na_2B_40_7. 3H_20$
E. $Na_2B_40_7. H_2O$

55.Outer transition elements belong to:
A. S-block
B. p-block
C. d-block
D. f-block
E. None of the above

56. Transition elements have coloured compounds because:
A. Their bond energy is low
B. they easily absorb energy
C. splitting of the five degenerated d-orbitals take place
D. d-orbitals are very close to p-orbitals
E. degenerate p-orbitals are present

57. In a double-bonded carbon atom (C=C):
A. hybridization occurs between the s-orbital and one p orbital
B. hybridization occurs between the s-orbital and two p- orbitals
C. hybridization occurs between the s-orbital and three p-orbitals
D. no hybridization occurs between the s- and p-orbitals E hybridization occurs between two s-orbitals and one p orbital

58. The radii of the second orbit of the hydrogen atom calculated from Bohr's model is:
A. 0.529 Aº
B. 4.8 Aº
C. 2.41 Aº
D. 3.4 Aº
E. 1 Aº

59. The amount of energy released by absorbing an electron in the valence shell is:
A. Ionization Energy
B. Electron Affinity
C. Electronegativity
D. Atomic Radius
E. Atomization Energy

60. Rate = K[N_2 O_5) has of reaction.
A. First order
B. Pseudo first order
C. Second order
D. Third order
E. Pseudo order

61. Which one of the following molecules has shortest distance of carbon atoms?
A. CH_3-CH_3
B. CH_2=CH_2
C. CH=CH
D. CH_3-CH_2- CH_3
E. CH_2=CH_2-CH_3

62. The most dense clement is:
A. Li
B. K
C. Ca
D. Ba
E. Rb

63. The isomers must have the same:
A. Physical properties
B. Molecular Formula
C. Structural Formula
D. Chemical properties.
E. Both B and C

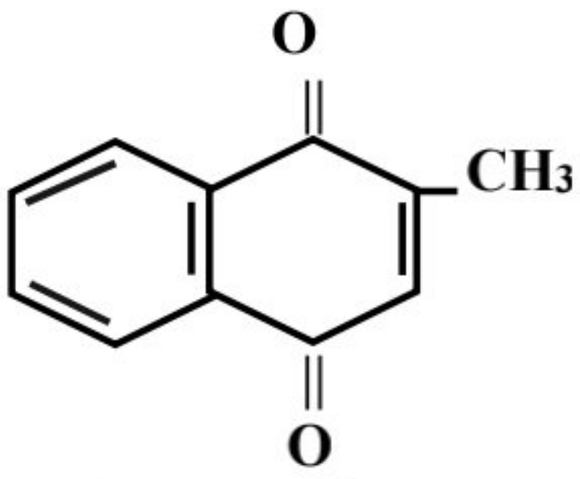
6-4. For a reaction 2A+B⇌C+D the active mass of B is kept constant and that of A is tripled. It is observed that the rate of reaction.
A. decreases three times.
B. decreases nine times

MDCAT Past Papers PRESTIGIOUS TESTING SERVICE 2000–2023

For Study Material Visit: Medicoengineer.com

C increases six times
D. increases nine times

65. Given below is a structure of:



A. Menadione
B. α-Tocopherol
C. Calciferol
D. Thiamine
E. Pyridoxine

66. In the final answer of the expression: -(29-20.2) (1.79 x 10^5)/1.37 the number of significant figures is:
A.1
B. 2
C. 3
D. 4

67. If we take 2.2 grams of CO_2, 6.02 x 10^{21} atoms of nitrogen and 0.03gram atoms of sulphur, then the molar ratio of C,N O and atoms will be:
A. 1:2:5
B. 5:1:2
C. 2:5:3
D. 5:1:3

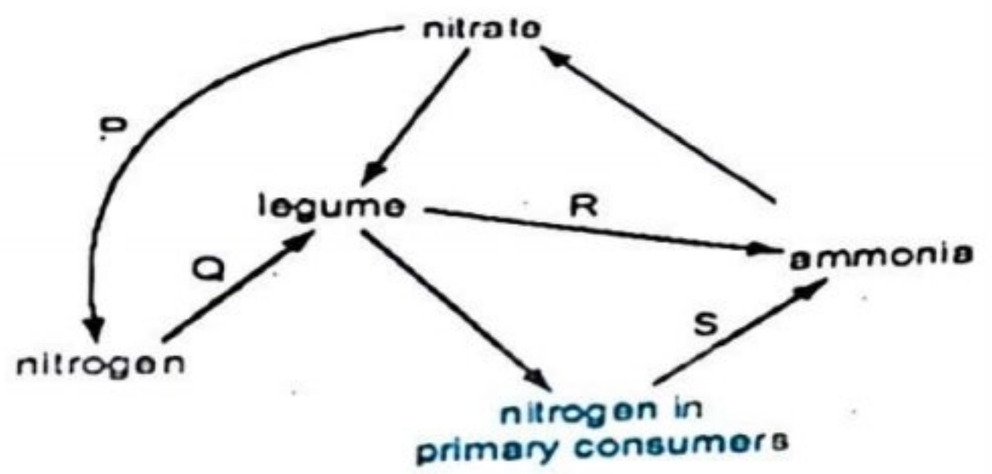
68. A system at equilibrium can be disturbed by:
A. Concentration change
B. Pressure change
C. Temperature
D. All of the above

69. Among the following electrons, which has highest energy?
A. n=3,l=2, m=0, S=+1/2
B. n =4,1=0, m= 0, S=1/2
C. n=3.1=1 m=1. S-1/2
D. n=3, I=0, m=0 S=-1/2

70. Equal weights of methane and hydrogen are mixed in an empty container at 25° C. The fraction of total pressure exerted by hydrogen is:
A 12
B. 8/9
C. 19
D. 16/17

BIOLOGY

71. The diagram shows a simplified nitrogen cycle.



Which row shows the correct labels for P, Q, R and S?

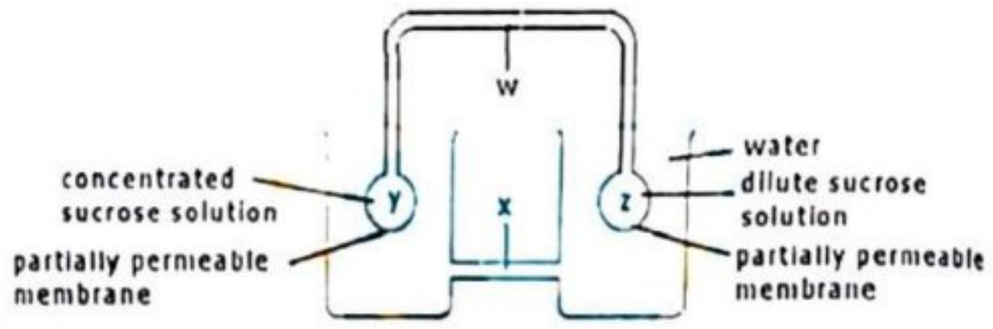
	P	Q	R	S
A	Denitrification by anaerobic bacteria	Nitrogen fixation nitrifying bacteria	Decay of leaf tissue by saprotrophic fungi	Ammonification by saprotrophic fungi
B	Lightening action on soil nitrates	Nitrogen fixation by nitrogen fixing bacteria	Decomposition using nitrogen enzyme	Decomposition by root nodule bacteria
C	Nitrification by anaerobic bacteria	Nitrification using nitrogenase enzyme	Decay of leaf tissue by saprotrophic fungi	Assimilation of organic nitrogen
D	Reduction by anaerobic bacteria	Nitrogen fixation by root nodule bacteria	Decomposition of organic nitrogen	Decay of urea by saprotrophic bacteria

72. In the commercial manufacture of Insulin, a human gene Is Inserted into which of these?
A. a chromosome of a human cell
B. a protein molecule in a yeast cell
C. the DNA of a bacterium
D. the nucleic acid in a virus

MDCAT
Past
BY PRESTIGIOUS TESTING SERVICE
2000-20

For Study Material Visit: Medicoengineer.com

73. The diagram shows a model to demonstrate the mass flow hypothesis of translocation.



In a plant, what are the structures W, X, Y and Z and what is the direction of flow of solution along W?

	Phloem	Xylem	Roots	Leaves	Direction to from among w
A	W	X	Y	Z	From z to y
B	W	X	Z	Y	From y to z
C	W	W	Y	Z	From y to z
D	W	W	Z	Y	From z to y

74. Many scientists believe that one of the following is/are evolutionary origin(s) of animals, plants and fungi?

A. Protists
B. Algae
C. Bacteria
D. Protozoans

75. In the human body, blood circulating from the gut to the heart passes through the:

A. aorta
B. kidneys
C. Ilver
D. lungs
E. spleen

76. The diagram shows the four types of human tooth



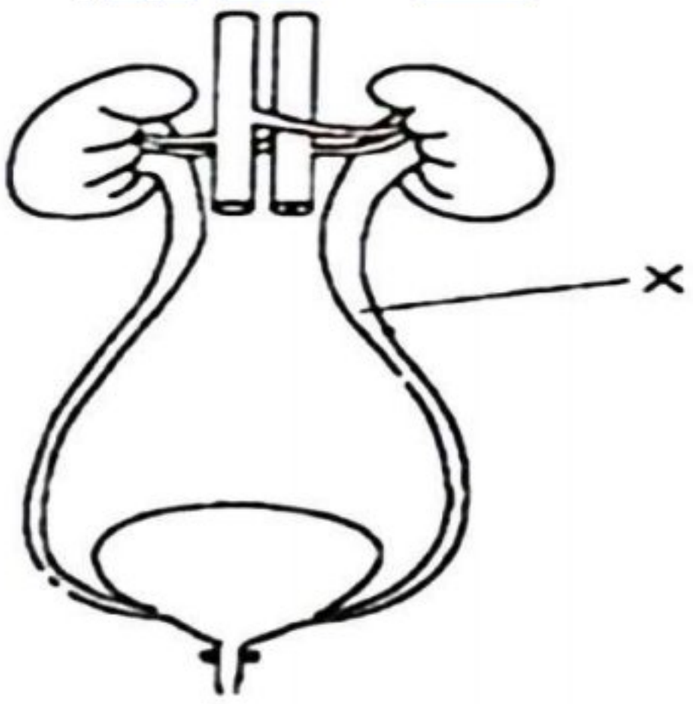
Which teeth are used for cutting rather than grinding food?

A. 1 and 2
B. 2 and 3
C. 3 and 4
D. 4 and 1

77. What are the functions of the inter, motor and sensory neurons in a reflex response?

	INTER NEURON	MOTOR NEURON	SENSORY NEURON
A	to connect neurons within the central nervous system	to conduct impulses to the effector from the central nervous system	To conduct impulses from the receptor to the central nervous system
B	To conduct impulses to the effector	To connect neurons within the central nervous system	To receive the stimulus
C	To conduct impulses from the central nervous system to the effector	To conduct impulses from the receptor to the central nervous system	To connect neurons within the central nervous system
D	To conduct impulses from the receptor to the central nervous system	To conduct impulses from the receptor to the central nervous system	To conduct impulses to the effector

78. The diagram shows the human urinary system.



Which substance is not found in the liquid at x In a healthy person?

A. glucose
B. Salt
C. toxins
D. water

79 Male and female sea urchina release their sperm and eggs Into the water where fertilization takes place. How can their reproduction be described?

A. asexual reproduction which results in genetically dissimilar offspring
B. asexual reproduction which results in genetically identical offspring

MD Past C BY PRESTIGIOUS TE 2000-20

For Study Material Visit: Medicoengineer.com

C. sexual reproduction which results in genetically dissimilar offspring
D. sexual reproduction which results in genetically identical offspring

80. Which vertebrate groups have scaly skin?
A. amphiblans and fish
B. amphiblans and mammals
C. fish and mammals
D. fish and reptiles

81. The following reaction occurs in the human alimentary canal, enzyme
starch

starch $\xrightarrow{\text{enzyme}}$ products
What is the enzyme and the product?

	ENZYME	PRODUCT
A	acid	glucose
B	alkall	energy
C	amylase	maltose
D	bile	amino acid

82.Archaeopteryx Is a transitional stage between the members of which one of the following pairs?
A. amphiblan... bird
B fish…. amphiblan
C. reptile... mammal
D. reptile bird
E mammal... man

83. In the Krebs cycle, substrate-level phosphorylation accompanies the formation of:
A. Citrate
B. Alpha-ketoglutarate
C. Succinate
D. Fumarate
E Oxaloacetate

84. When a physician elicits the knee-jerx reflex by tapping deep tendons in the knee, the normal response is for the lower leg to swing forward. When this happens:
A. Muscles in the front of the thigh are contracting and muscles in the back of the thigh are relaxing
B. Muscles in the front of the lower leg are contracting and muscles in lower leg are relaxing
the back of the
C. Muscles in the back of the thigh are contracting and muscles in the front of the thigh are relaxing
D. Muscles In the back of the lower leg are contracting and muscles In the front of the lower leg are relaxing
E None of the above

85.of the following, which is the incorrectly paired one?
A. Robert Hooke cell wall
B. Schleiden and schwann... cell theory
C. Robert Brown... nucleus
D. Watson and Crick... DNA model
E. Virchow mosaic model of plasma membrane

86 Identify the phylum in which the larva is bilaterally symmetrical but the adult is radially symmetrical:
A. Ctenophora
B. Coelentreta
C. Echinodermata
D. Sipunculoidea

87. The botanical name of gum tree is:
A. Acacia nilotica
B. Mimosa pudica
C. Acacia catechu
D. Prosopis glandulosa
E. Albizzia Lebbek

88.A pure-breeding plant with the dominant phenotype of character Pand the recessive phenotype of character Q

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

For Study Material Visit: Medicoengineer.com

was crossed with another pure-breeding plant with the recessive phenotype of character P and the dominant phenotype of Q. The offspring of this cross were crossed with a double homozygous recessive for Pand Q and the following results obtained: 22 were phenotypically dominant for P and recessive for O. 5 were phenotypically dominant for both P and Q 4 were phenotypically recessive for both P and Q. 24 were phenotypically recessive for P and dominant for Q. Which one of the following types of inheritance is Illustrated by these results?

A. gene linkage of P and Q
B. independent segregation of P and Q
C. Mendellan dlhybrid Inheritance
D. multiple alleles
E polygenic Inheritance

89. How many metacarpals are present in the hand?

A. 4
B. 3
C. 6
D. 5
E. 8

90, which of the following is NOT a difference that would allow one to distinguish between a prokaryotic and a gukaryotic coll?

I. Presence or absence of the nucleus
II. Presence or absence of the cell wall
III. Membrane-bound versus no membranous-bound organelles

A. I only
B. II only
C.III only
D. I and II only
E. I. II and III

91. Some enzymes require the presence of a nonprotein r catalytically. An enzyme devold of this molecule is called

A. holoenzyme
B apoenzyme
C coenzyme
D zymoenzyme

92. The events shown below occur during different phases of mitosis:

I. spiralization of DNA
II. hydration of DNA
III. centromeres split
IV. centromeres attach to spindle fibres
V. DNA replicates

Which one of the following correctly identifies each of the phases described

	Inter phase	prop hase	meta phase	anap hase	telop hase
A	I	II	III	IV	V
B	I	V	IV	II	III
C	V	I	IV	III	II
D	II	IV	I	III	V
E	V	IV	I	II	III

93. When a fetus is in the uterus, what carries oxygen away from the placenta?

A. The amniotic fluid.
B. The amniotic sac
C. The lining of the uterus
D. The umbilical cord

94. The floral formula of family mimosaceae Is:

A. $\oplus, \male, K_{(5)}, \overbrace{C_{(5)}\ A_5}, \underline{G}_{(2)}$
B. $+, \male, K_{(5)}, C_{1+2+(2)}, A_{(9)+1}, \underline{G}_1$
C. $+, \male, K_{(5)\text{ or }5}, C_5, A_{10}, \underline{G}_1$
D. $\oplus, \male, K_{(5)}, C_{5\text{or}(5)}, A_{\infty\text{ or }(10)}, \underline{G}$
E. None of the above

95. Which type of protein structure contains the three dimensional structure?

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

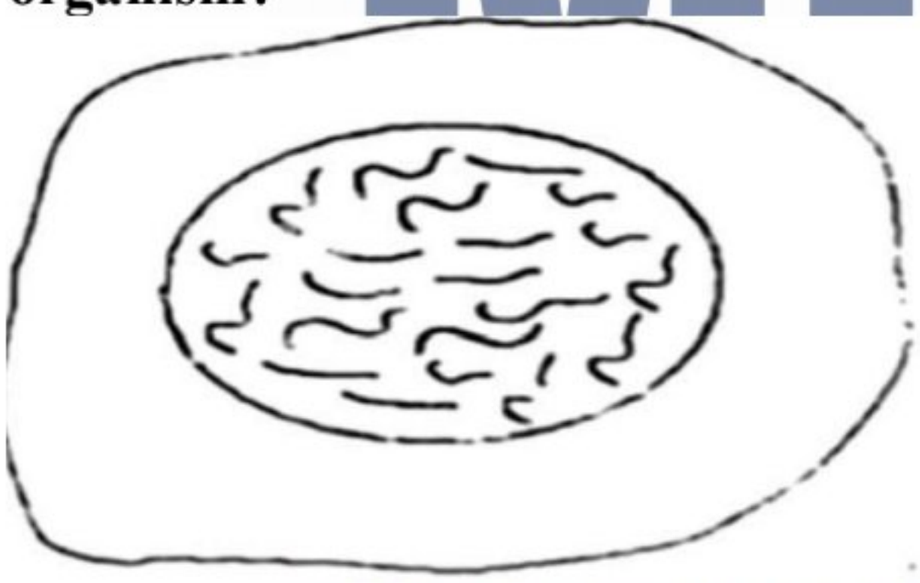
For Study Material Visit: Medicoengineer.com

A. primary
B. secondary
C. tertiary
D. quaternary

96. Which of the following describes the movements involved in breathing out?

	movements of ribs	movements of diaphragm
A	down and in	downwards
B	down and In	upwards
C	up and out	downwards
D	up and out	Upwards

97. The diagram shows a cell of an organism formed by reduction division What is the diploid number for this organism?



A. 10
B. 20
C. 30
D. 40
E. 44

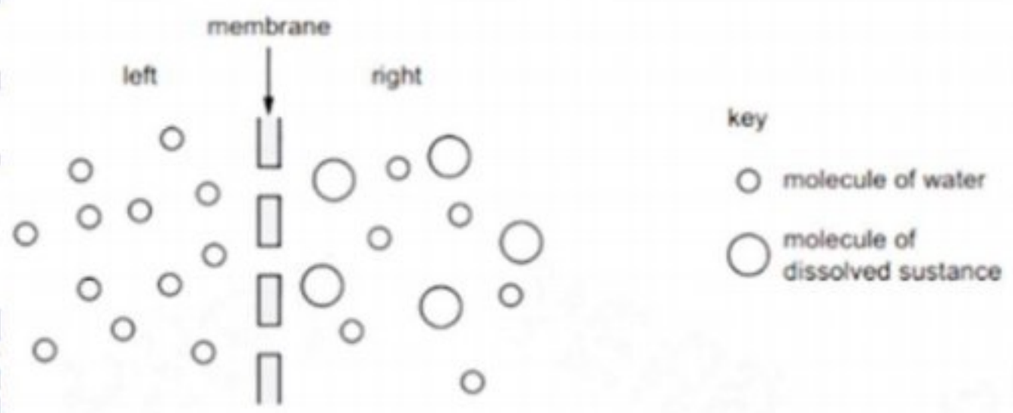
98. The diagram shows a metabolic pathway.

Reactant —Enzyme 1→ Substance X —Enzyme 2→ End Product
Substance Y —Enzyme 3→

What would be the effect of adding a small amount of a non competitive inhibitor of enzyme 2?
A. Enzyme 2 would be partially denatured.
B. Substance X would increase in concentration.
C. Substance Y would no longer be formed.
D. The initial reactant would no longer be metabolized.
E. The effect would be negligible.

99. Which processes are essential in making nitrogen in dead plant material available to growing plants?
I. ammonification
II. deamination
III. nitrification
IV. nitrogen fixation
A. I, II and III only
B. I. II and IV only
CI. III and IV only
D. II. III and IV only

100. The diagram represents two liquids, separated by a membrane through which osmosis can occur.



What movement of molecules will occur?
A. Molecules of dissolved substance move from left to right.
B. Molecules of dissolved substance move from right to left.
C. Overall, water molecules move from left to right.
D. Overall, water molecules move from right to left.
E None of the above

MDCAT
BY PRESTIGIOUS TE
2000-2018

For Study Material Visit: Medicoengineer.com

ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	A	26	E	51		76	A
2	A	27	C	52	A	77	A
3	E	28	B	53	CA	78	A
4	D	29	C	54	A	79	C
5	E	30	A	55	C	80	D
6	A	31	A	56	C	81	C
7	D	32	D	57	B	82	D
8	A	33	D	58	C	83	C
9	D	34	C	59	B	84	A
10	C	35	A	60	A	85	E
11	D	36	B	61	C	86	C
12	A	37	D	62	D	87	A
13	A	38	C	63	B	88	A
14	E	39		64	D	89	D
15	A	40	C	65	A	90	B
16	C	41	C	66	C	91	B
17	A	42	B	67	D	92	C
18	B	43	A	68	D	93	D
19	D	44	D	69	A	94	D
20	B	45	B	70	C	95	C
21	A	46	A	71	A	96	B
22	B	47	D	72	C	97	E
23	A	48	D	73	B	98	E
24	A	49		74	B	99	C
25	C	50	E	75	C	100	C

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023